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import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
import matplotlib.pyplot as plt

url = r"DATA SET PATH SMSspambase.data"

df = pd.read_csv(url, sep='\t', names=['label', 'message'])

print(f"Total: {len(df)} | Ham: {sum(df['label']=='ham')} | Spam:
{sum(df['label']=='spam')}")

df['label_num'] = df['label'].map({'ham': 0, 'spam': 1})

X_train, X_test, y_train, y_test = train_test_split(
    df['message'], df['label_num'], test_size=0.2, random_state=42
)

tfidf = TfidfVectorizer(max_features=5000, stop_words='english')
X_train_vec = tfidf.fit_transform(X_train)
X_test_vec = tfidf.transform(X_test)

model = MultinomialNB()
model.fit(X_train_vec, y_train)

y_pred = model.predict(X_test_vec)
print(f"\nAccuracy: {(y_pred == y_test).mean():.2%}")

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

df['label'].value_counts().plot(kind='pie', autopct='%1.1f%%',
ax=axes[0])
axes[0].set_title('Ham vs Spam')

feature_names = tfidf.get_feature_names_out()
spam_score = model.feature_log_prob_[1, :] - model.feature_log_prob_[0,
:]
top_idx = spam_score.argsort()[-10:]

axes[1].barh(range(10), spam_score[top_idx])
axes[1].set_yticks(range(10))
axes[1].set_yticklabels([feature_names[i] for i in top_idx])
axes[1].set_title('Top 10 Spam Words')

plt.tight_layout()
plt.show()

test_messages = [
    "Free prize! Claim now!",
    "Hey, meeting at 3pm tomorrow",
    "You won $1000! Click here"
]
```

```
for msg in test_messages:
    vec = tfidf.transform([msg])
    pred = model.predict(vec)[0]
    print(f"\n'{msg[:30]}...' → {'SPAM' if pred==1 else 'HAM'}")
```